# Mini Case Study: Physical Data Visualization

*Football Data Analytics*

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## Overview

You are provided with physical performance data collected from a training session of a U20 Bangladesh national football team player. Your task is to explore this dataset, produce a set of meaningful visualizations, and accompany each plot with your own written interpretation of what the data reveals about the player's performance.

## Learning Objectives

- Apply data analysis techniques to a real-world sports performance dataset.
- Design and implement clear, informative visualizations using Python (Matplotlib, Seaborn) or comparable tools.
- Interpret physical metrics such as speed, heart rate, acceleration, body rotation, and PlayerLoad in the context of football training.
- Identify and communicate correlations between metrics to derive actionable insights.

## Required Visualizations

At minimum, your submission should reproduce the following visualizations (sample plots are provided for reference):

1. **GPS positional data —** pitch coverage track colored by speed zone, plus a position heatmap.
2. **Speed and heart rate profile —** time series with walking, jogging, and running zones highlighted.
3. **Body rotation, sprint events, and cumulative counts —** angular velocity over time, distribution of body rotation, cumulative event counts (acceleration, deceleration, sprint), and a speed vs body rotation scatter plot.
4. **Acceleration profile and PlayerLoad —** instantaneous acceleration, acceleration distribution, cumulative PlayerLoad, and rolling PlayerLoad intensity.

## Additional Analysis

Going beyond the required plots will improve your grade. You are encouraged to explore correlations between metrics and present any insights that the basic visualizations do not capture. Examples include:

- Correlation matrices or pairwise scatter plots (e.g., speed vs heart rate, acceleration vs PlayerLoad).
- Heart rate recovery analysis between high-intensity bursts.
- Fatigue indicators across the session (e.g., decline in peak speed or sprint frequency over time).
- Zone-based time distribution (percentage of session spent in each speed or HR zone).

## Insights and Commentary

Each visualization must be accompanied by a short written commentary (3–6 sentences) explaining what the plot shows and what it tells you about the player's physical performance. Commentary must be written in your own words. You should be prepared to defend your interpretations during the viva; if you cannot explain your own analysis, you may receive zero marks for the entire case study.

## Submission Requirements

- A single Jupyter Notebook (.ipynb) or PDF report containing your code, plots, and commentary.
- All plots must be clearly labelled (titles, axes, units, legends) and reproducible from your code.
- Commentary must be in your own words. Plagiarized or AI-generated text presented as your own will result in zero marks.

## Grading

This case study contributes 5% to your total course grade, distributed approximately as:

- Required visualizations (correctness and clarity): 50%
- Insights and commentary: 25%
- Additional analysis and correlations: 15%
- Overall report quality: 10%

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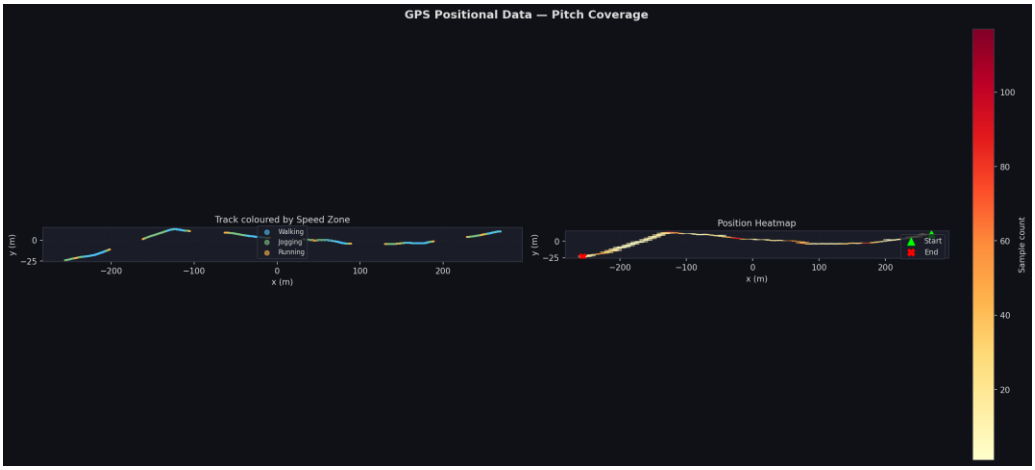
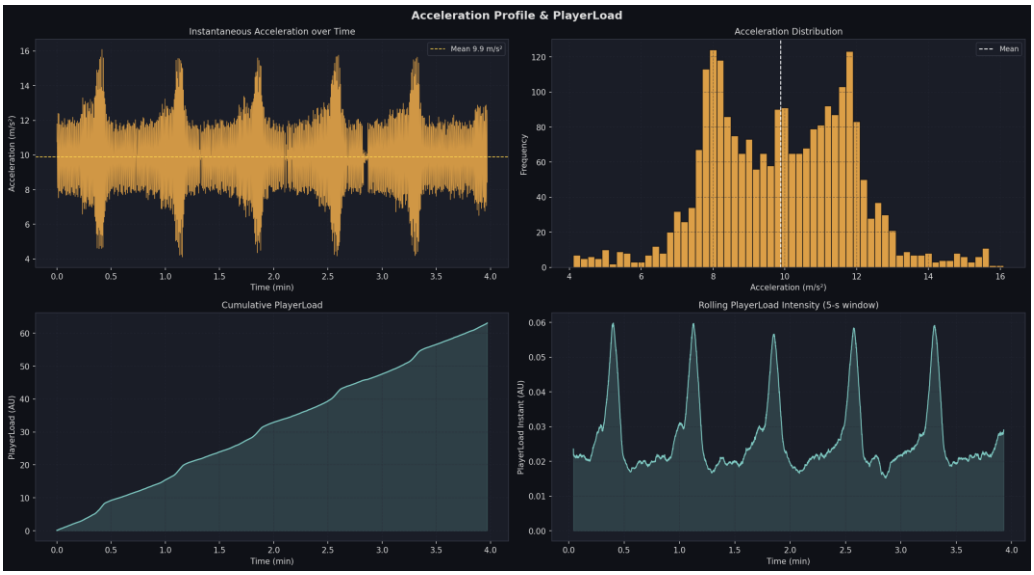
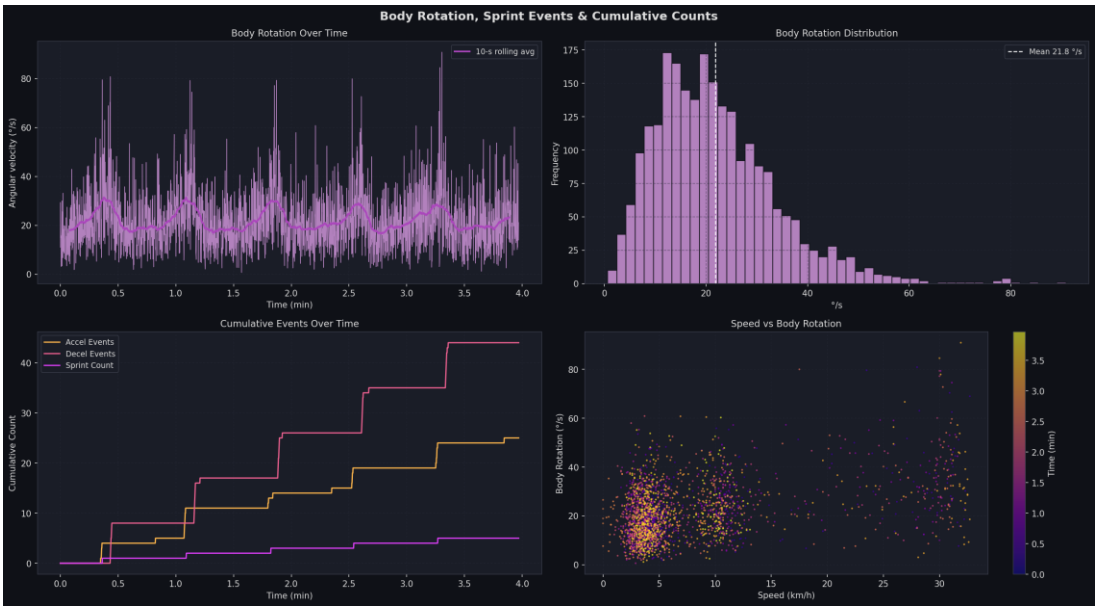
**Note:** Sample visualizations given below from the original case study brief are intended only as a guide. You are encouraged to improve on them with better styling, additional context, and deeper analytical depth.

## Sample Visualization



Speed & Heart Rate Profile
Walking
Jogging
Running
Speed (km/h)
HR
Avg 132 bpm
Heart Rate (BPM)
Elapsed Time (min)


Acceleration Profile & PlayerLoad
Instantaneous Acceleration over Time
Mean 9.9 m/s²
Acceleration (m/s²)
Time (min)
Acceleration Distribution
Mean
Frequency
Acceleration (m/s²)
Cumulative PlayerLoad
PlayerLoad (AU)
Time (min)
Rolling PlayerLoad Intensity (5-s window)
PlayerLoad Instant (AU)
Time (min)


Body Rotation, Sprint Events & Cumulative Counts
Body Rotation Over Time
10-s rolling avg
Angular velocity (°/s)
Time (min)
Body Rotation Distribution
Mean 21.8 °/s
Frequency
°/s
Cumulative Events Over Time
Accel Events
Decel Events
Sprint Count
Cumulative Count
Time (min)
Speed vs Body Rotation
Body Rotation (°/s)
Speed (km/h)
Time (min)